How Isaacus teaches vector embeddings

An analysis of the sequence, examples and reader controls in Abdur-Rahman Butler's interactive guide.

The crucial move happens in a table

The author gives warm and muggy the same temperature value. A reader can see the problem immediately: the words feel different, yet the numbers say they are alike. Humidity now has a job to do. By the time the author introduces another dimension, the reader already knows why one is needed.

This is the feature most worth replicating. Give readers a judgement they can make with knowledge they already have. Let them discover exactly where that knowledge runs out. Introduce the technical idea as a way to resolve that difficulty.

What the interaction contributes

A control earns its place when the reader can predict what changing it will do. Choosing another reference sentence tests which meaning of a word matters. Raising a matching threshold tests how many records will still be grouped. Both actions let someone disagree with the explanation, try it and inspect the result.

Where the guide asks more of its audience

The opening game asks the reader to learn the controls, find suitable words and enclose a target. Someone can improve at that task without being able to explain a similarity score. The game offers a reason to explore. The sentence comparison gives a much clearer indication of what the reader understands.

Scope and evidence

I reviewed the article in reading order and inspected its desktop interface on 6 November 2026. I changed the sentence selector and duplicate threshold. The similarity plot appeared blank in both display modes in this browser, so its numerical examples below come from the accompanying prose. No reader testing, mobile testing or accessibility conformance audit was conducted.

Primary source: Abdur-Rahman Butler, Isaacus, 14 October 2026.

How one idea earns the next

Each stage changes the question a reader can answer. The order matters because the same idea is put to work again later.

1 Try the outcome	A word-map game gives the reader something to do before the terminology arrives. Revisiting it later gives the reader a second chance to explain a familiar object.
2 Build a simple model	Character codes establish that symbols can have numeric representations. The temperature scale goes a step further: numbers can preserve a relationship people recognise. The warm/muggy tie exposes what that scale loses.
3 Move from a list to a place	A selected word, its numeric coordinates and a projected view appear together. The numbers establish what the model produces. The plot gives a reader a way to talk about relationships. The text explains that flattening the space distorts distances.
4 Move between places	An analogy supplies a worked example of moving in a direction. An expression field and suggested variations let the reader try the operation again. Approximate answers and the author's caveat put a limit on the analogy.
5 Compare a pair, then a map	Sentences give the reader a reason to compare two directions with one score. Returning to the large word map then poses a different question: which relationships survive when thousands of words share a flat display?
6 Use the comparison	Chronos, a fictional time-travel business, supplies a continuing task. A query finds an article, a question finds a passage at the right level, a ticket finds a team and issue reports find one another. What changes is the thing being compared.
7 Return to the opening	Grouping issue reports explains how regions of the opening map acquired labels. A discussion of responsibility then moves from what a system can group to what someone should do with those groups.

Visit the temperature example, word map or Chronos case study.

What the controls actually teach

Change a sentence and hold the alternatives steady

The sentence selector varies one thing: the text used as the reference. The other sentences are familiar by the second comparison. That makes it possible to ask whether shared words or shared meaning better explain the ordering.

Animal reference	The article reports 0.55 for a flying fox leaving a grotto and 0.48 for a sports bat leaving someone's hands, each compared with a bat leaving a cave. The first description changes the nouns. The second repeats conspicuous words while changing their meaning.
Sport reference	With the sports bat as reference, a sentence about losing one's grip on a racket scores 0.60. The reader can account for the change using ordinary language before interpreting the geometry.

The small 0.07 gap in the first comparison is instructive. It supports a claim about the order of these examples. It supplies no general pass mark for "same meaning". A useful prompt before revealing the scores would be: "Which sentence would you expect to come closer, and which word might mislead you?"

Source of scores: Semantic similarity.

Change a rule and watch the same reports regroup

The duplicate-report example keeps the eleven issue reports fixed. I raised the minimum similarity required for a connection. The interface produced these outcomes:

53% threshold	Three groups contained three, three and two reports respectively. Three reports stood alone.
90% threshold	All eleven stood alone. The three earlier groups disappeared.

A stricter rule can separate reports about the same fault. The rule changed; the faults and reports did not. That is a more useful lesson than memorising a preferred setting. Opening individual reports also gives the reader evidence with which to challenge a proposed match.

Direct check: Clustering and deduplication, 6 November 2026. Percentages are displayed similarity thresholds.

What to keep, and where to adapt

Write the sentences around the control

Before the sentence selector, the author names the action and tells the reader what to compare. Below it, the prose interprets two particular outcomes. The reader has a route into the figure and a way to check their reading afterwards. In the desktop layout, a narrow reading column keeps the prompt close to the control, while a contents list gives readers a route back to an earlier concept.

Choose examples that expose a competing explanation

The bat example makes word overlap a plausible but imperfect rule. The two search panels do something similar: both get the same customer query and article collection, so readers can inspect why the top results differ. Isaacus supplies its own model and a simple keyword baseline. Those choices support a demonstration. A claim about typical search performance would need representative queries and independent relevance judgements.

Keep the example familiar as the question gets harder

Chronos is an unusual business with ordinary problems: payments fail, customers need help and staff triage tickets. Reusing that setting lets an author change the operation without repeatedly introducing new characters and stakes. The long code samples offer a route to implementation. For a reader following only the concept, placing that code behind an optional disclosure would shorten the distance between question and demonstration.

Show where a picture can mislead

Nearby dots on a flat map invite a stronger claim than the projection supports. The guide explains the loss of information and distinguishes the display from the vectors used for calculations. Carry that distinction into the widget itself. Give the selected pair a text readout, label the space used for its score, and make the result available when the plot fails to draw.

Give play and assessment different jobs

A small loop in the game rewards spatial navigation and vocabulary as well as semantic judgement. Keep it as an invitation to explore. To assess understanding, give the reader an unfamiliar sentence pair, ask for a ranking and listen to the reason. Similarly, a displayed similarity of 52% can look like a probability of being correct. Explain the score beside the result, especially when a correct/incorrect badge is also visible.

Examples: semantic search comparison and projected word map.

A storyboard you can test

Worked example for an evidence atlas: help a reader see why the area with the most filings can also have the lower filing rate. All figures below are invented for teaching.

1 Ask for a judgement	Show Area A with 120 filings and Area B with 60. Ask which recorded more filings. The answer is A. Keep those figures on screen when the question changes.
2 Expose what is missing	Ask which area recorded more filings per 100 businesses. Invite the reader to say what else they need. Then reveal 2,000 businesses in A and 500 in B.
3 Let one action test it	Offer a switch between total filings and filings per 100 businesses. Retain the area labels and raw figures. The rates are 6 for A and 12 for B. Show the calculation beside each result and provide the same information in text.
4 Explain the reversal	A recorded twice as many filings across four times as many businesses. Its filing rate was therefore half B's. This measure counts filings. One business can file several times, so the share of businesses that filed is a separate question.
5 Try a fresh case	Give Area C 90 filings and 1,500 businesses. Ask the reader to compare C with A before showing the chart. C has fewer filings and the same rate. A tie checks whether the reader learned the principle or simply expects a reversal.

How to tell whether interaction helped

Compare two versions with the same words, numbers and examples: one with the switch and one showing both states as a static table. Assign readers from the intended audience to a version. Ask both groups to solve the fresh case unaided. Score the answer and the explanation separately. Record whether they confuse filings with businesses that filed. Clicks and completion time can help locate friction; success means applying the distinction to new data.

There is a reason to test this sequence. In three classroom studies, Schwartz and Bransford found that students who analysed contrasting cases and then received an explanation made more accurate predictions about a new case a week later than comparison groups. That finding supports testing the design in this audience. Its effect here remains an open question.

Research: Schwartz & Bransford (1998), A Time for Telling, Cognition and Instruction, 16(4), 475–522.